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Irregular sleep/wake patterns in student-athletes exposed to early morning training

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ABSTRACT

This study aimed to examine the sleep parameters and sleep/wake regularity of a cohort of student-athletes who start training between 06:30 and 07:00. Twenty-one male Rugby Union players, aged 21 ± 2 years and competing at a national level, were assessed using actigraphy over two weeks, and the Athlete Sleep Screening Questionnaire (ASSQ). Sleep/wake regularity was calculated using the Sleep Regularity Index (SRI). Wilcoxon signed-rank tests showed that nocturnal sleep preceding morning training had a significantly shorter sleep duration (1.8 hr, $r = .67$), and advanced sleep onset (0.9 hr, $r = .50$) and sleep offset times (3.2 hr, $r = .85$) compared to nocturnal sleep preceding free days. The variability of training demands resulted in an inconsistent sleep pattern between consecutive days, resulting in a median SRI score of 67.0 (interquartile range: 17.0). Pearson correlations revealed that lower SRI was significantly associated with a higher daily sleep duration including naps ($r = -.62$), delayed sleep onset ($r = -.50$) and sleep offset ($r = -.60$), and a later chronotype assessed using the ASSQ ($r = .52$). These findings indicate that early morning training is a factor contributing to irregular sleep/wake patterns in student-athletes, and where feasible should be scheduled at an alternative time.

KEYWORDS

Athletes; chronotype; sleep; sleep regularity; students

Introduction

Sleep has garnered growing attention within student-athlete populations, with the National Collegiate Athletic Association (NCAA) establishing a task force to review current evidence and shape future practices to support healthy sleep habits (Kroshus et al., 2019). Previous research has indicated that student-athletes exhibit a high prevalence of undesired sleep characteristics, including poor perceived sleep quality (Leduc et al., 2020; Mah et al., 2018; Stephenson et al., 2022), sleep durations under the public health recommendations of seven to nine hours per night for young adults (Carter et al., 2020; Driller et al., 2017; Hirshkowitz et al., 2015; Mah et al., 2018), and poor sleep hygiene behaviours (Driller et al., 2017). In addition, about one-quarter of sleep screened student-athletes present with moderate to severe sleep disturbances (Rabin et al., 2020; Rebello et al., 2022). The reasons underpinning these findings are diverse, with both sport-related risk factors, such as travel to and from competition (Walsh et al., 2021), and academic-related risk factors, such as emotional and academic stress (Lund et al., 2010). Accordingly, student-athletes should be considered a unique population within sleep research.

The scheduling of training, particularly when placed early in the morning, constitutes a significant sleep risk factor in athletes (Walsh et al., 2021). In elite athletes, an advance in morning training timing significantly reduces total sleep time the prior night (Sargent et al., 2014), with this finding supported by subsequent research on student-athletes (Benjamin et al., 2019; Merfeld et al., 2022). The scheduling of training early in the morning is common amongst student-athletes and stems from the logistical necessity of fitting training around academic teaching hours. However, this timing conflicts with the changes in chronotype that occur during young adulthood, marked by a peak phase delay at this age (Roenneberg et al., 2004). Consequently, adapting to early morning starts may pose particular challenges for student-athletes. Furthermore, it has been shown that athletes are more likely to participate in sports that align with their chronotype, and the misalignment in chronotype and training timing in student-athletes could result in challenges adapting to this schedule and hinder progression to higher performance levels in the future (Lastella et al., 2016).

Walsh et al. (2021) have noted that the timing of training and competition can disrupt consistent sleep patterns in performance athletes. Maintaining a consistent sleep/wake pattern is an important component of healthy sleep to support the circadian process of sleep regulation, with a recent consensus statement indicating that regular sleep timing is important for various health and performance outcomes (Sletten et al., 2023). Additionally, irregular sleep/wake patterns have been associated with poorer academic attainment (Phillips et al., 2017). Previous sleep research in athletes has often presented sleep parameters as an average value over the monitoring period without reporting any measure of within-person variability between days (Halson et al., 2022). A recent systematic review identified only 16 studies reporting within-person variability in sleep for athletes (Kemp et al., 2023). The most common metric used in these studies is intra-individual variability, where sleep parameters on each night are compared to an individual's average sleep pattern (Fischer et al., 2021), and it is evident that athletes exhibit high intra-individual

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variability in sleep parameters (Kemp et al., 2023; Nedelec et al., 2018). A study on British student-athletes revealed greater intra-individual variability in sleep parameters compared to non-athlete students (Leduc et al., 2020), while elite junior athletes exhibited greater intra-individual variability in time in bed and sleep duration measures compared to elite and sub-elite senior athletes (Caia et al., 2017). Therefore, student-athletes may be more susceptible to greater sleep variability than other athlete groups.

One potential limitation in applying intra-individual variability to student-athlete groups is that it functions as an 'overall' metric, establishing variability in comparison to an individual mean across a monitoring period. To gauge day-to-day variability, a 'consecutive' measure is required that may be more reflective of circadian disruption (Fischer et al., 2021). This is especially pertinent for a student-athlete population, where significant variations in daily schedules may influence sleep parameters between consecutive days. The Sleep Regularity Index (SRI) is a novel consecutive metric that calculates the probability of an individual being in the same sleep/wake state at time points 24 hours apart (Phillips et al., 2017). Another advantage of assessing sleep across a 24-hour window is the ability to capture daytime napping, which is a common practice amongst student-athletes with up to 80% reporting napping at least once per week (Mah et al., 2018; Stephenson et al., 2022), and often employed by athletes to supplement lost nocturnal sleep (Walsh et al., 2021).

The SRI has previously been used to evaluate sleep regularity in University students (Phillips et al., 2017; Windred, Stone, et al., 2021) and elite athletes (Alves Facundo et al., 2022; Halson et al., 2022; Teece et al., 2024). Understanding the sleep/wake patterns of student-athletes has important implications for sport stakeholders, and can help make informed decisions on the timing of training sessions to facilitate healthy sleep practices, including regular sleep/wake timing. However, despite a rationale to anticipate irregular sleep/wake patterns in student-athletes, it has not been examined using a consecutive measure of sleep regularity in empirical research. Therefore, this study aimed to; (1) present the sleep parameters (including daytime napping) and SRI of a cohort of student-athletes, (2) examine differences in sleep parameters when categorised by day of the week and the presence or absence of morning training, and (3) examine the relationship between SRI and sleep parameters, including napping.

Methods

Participants

Twenty-eight male student-athletes were recruited from a single Rugby Union team competing in the British Universities and Colleges Sport Super Rugby league, the highest level of university competition in the United Kingdom, and gave informed consent to participate in the study. Student-athletes were excluded from participation if they had any previous history of a sleep-related disorder, current medication that may interfere with sleep, or any concussion since the start of the competitive season. These were self-reported and checked against medical records kept by team staff. Participants were required to provide valid actigraphy data for ≥5 days of overlapping epochs of actigraphy to enable an accurate assessment of sleep regularity for inclusion (Fischer et al., 2021). Six participants were excluded for insufficient valid actigraphy data, one of which was related to a device fault. One further participant was excluded for a suspected irregular sleep wake rhythms disorder (SRI = 3.9 with multiple sleep bouts and fragmented nocturnal sleep), with the participant signposted towards their General Practitioner.

Twenty-one participants were included in statistical analyses with a mean (±SD) age of 21 ± 2 years, height of 1.80 ± 0.17 m, and mass of 97.9 ± 24.5 kg. The majority of the sample identified as White ($n = 18$), while three participants identified as Black, Black British, Caribbean, or African. The level of study varied across undergraduate 1st year ($n = 7$), 2nd year ($n = 6$), 3rd year ($n = 5$) and postgraduate ($n = 3$), with all participants residing off-campus. The sample included both forwards ($n = 11$) and backs ($n = 10$) playing positions.

This study expanded upon a pilot study with 10 participants that are not included in this dataset. A conference proceeding summarising the pilot has been previously published (S. Wilson et al., 2023). The study received approval from Hartpury University Ethics Committee (ETHICS2023-01).Participants were offered individual feedback and entered into a prize draw for three £25 gift vouchers upon study completion.

Procedure

Data collection was conducted between 9 October 2023, and 23 October 2023, during the competitive sporting season and academic teaching weeks. The training and match schedule during the data collection period is detailed in Table 1. Recovery sessions on Thursday were optional, and Saturday conditioning sessions were completed at a self-selected time

Table 1. Training and match schedule of participants split by day.

Day of week	Session type	Timing
Monday	Field and gym training	Week 1 and Week 2: 06:30–10:30 and 12:00–12:30
Tuesday	Field training	Week 1: 06:45–08:30
		Week 2: 06:30–08:30
Wednesday	Match (including preparation and travel)	Week 1: 14:00–22:00 (away, kick-off 18:00)
		Week 2: 17:15–21:00 (home, kick-off 18:30)
Thursday	Recovery – Medical and Yoga	Self-scheduled
Friday	Field and gym training	Week 1: 06:30–10:30
		Week 2: 07:00–10:30
Saturday	Conditioning	Self-scheduled
Sunday	Rest day	–

with an approximate duration of 1.5 hr. All other sessions were mandatory. The academic timetable commenced no earlier than 09:30 with no lessons scheduled on the Wednesday before matches.

Participants were instructed to maintain their usual sleep habits throughout the study. On each morning, participants completed a short questionnaire including a shortened version of the core Consensus Sleep Diary (Carney et al., 2012), and self-reported training attendance and daytime napping. This was administered in both paper and online formats based on participant preference.

Measures

Before data collection, participants completed the Athlete Sleep Screening Questionnaire (ASSQ) (Samuels et al., 2016). The ASSQ is a validated sleep screening tool for athlete populations (Bender et al., 2018; Samuels et al., 2016). Six Likert scale questions are summed to provide a sleep difficulty score (ASSQ-SDS), and can be used to categorise athletes into none (0–4), mild (5–7), moderate (8–10) and severe (11–17) sleep problem groups. Chronotype is estimated through four Likert scale questions that are summed (ASSQ-Chronotype), and range from 0 (greatest eveningness) to 14 (greatest morningness). ASSQ-Chronotype does not cluster participants into chronotype groups, but can be used to identify extreme evening-types (ASSQ-Chronotype ≤4). The ASSQ includes additional modifier questions related to sleep-disordered breathing, travel-related disturbance, napping, caffeine consumption and electronics use which were not used for analyses.

Actigraphy-derived sleep parameters were collected using GENEActiv monitors (Activinsights, Cambridge, United Kingdom). These record movement through a tri-axial accelerometer with a range of ± 8 *g*, light wavelength (range: 400–1100 nm), light intensity (range: 0–3000 lux), and temperature (range: 0–60 °C). Monitors sampled at 40 Hz to allow sufficient storage capacity over the data collection period. Participants were asked to wear monitors on the non-dominant at all times except for contact training and to use the event marker button to denote sleep onset and offset. Data was extracted and saved in raw format files (.bin) using GENEActiv PC software v3.3.

The GGIR v3.0 R-package was used to process raw accelerometer data in RStudio (Migueles et al., 2019). Initial signal processing included automatic calibration using local *g* as a reference, detection of abnormal values, non-wear detection, and the quantification of dynamic wrist acceleration using the Euclidean Norm Minus One metric in 5-second epochs (Van Hees et al., 2013, 2015). The Sleep Period Time (SPT) was determined using a pre-defined analysis pathway incorporating event markers, raw accelerometer and light data, and/or sleep diary entries. The full analysis pathway is presented in the Supplementary File (Figure S1). This process differentiated sustained inactivity bouts (SIBs) classified as sleep from sedentary periods outside the SPT window. SIB identification utilised the Van Hees et al. (2015) algorithm, which classifies periods where wrist rotation (z-axis) does not change by > 5° for a minimum of 5 minutes. The algorithm has been validated using GENEActiv monitors with an overall accuracy of 83% (sensitivity: 91%, specificity: 45%) compared to polysomnography (Van Hees et al., 2015). Subsequent research has indicated comparable outcomes to other research-grade actigraphy monitors (Plekhanova et al., 2020). The same analysis strategy was applied to detect daytime napping, with the SPT for a given nap set using sleep diary entries.

The actigraphy-derived sleep parameters examined for each sleep episode (including naps) were: *sleep onset* (hh:mm), the first instance of sleep in the SPT; *sleep offset* (hh:mm), the final instance of sleep in the SPT; and *Total Sleep Time* (TST; hr), the total amount of sleep within the SPT. In addition, the following parameters were extracted for nocturnal sleep episodes: *sleep period* (hr), the time between sleep onset and sleep offset; *Wake After Sleep Onset* (WASO, hr), the time awake between sleep onset and sleep offset; *sleep efficiency* (SE, %), the proportion of the sleep period spent asleep (i.e. TST/sleep period); and *Sleep Onset Latency* (SOL, hr), the period of time between reported bedtime and sleep onset. The *cumulative sleep duration* across each 24-hour from 12:00 to 12:00 was calculated by adding the TST of all sleep episodes including naps.

Data processing

The Sleep Regularity Index (SRI) was used as a measure of the regularity of sleep/wake patterns (Phillips et al., 2017). The SRI calculates the probability of being in the same sleep/wake state at any given time point 24-hours apart and averaged across the study duration. SRI is scaled to a theoretical range between −100–100, but in practice typically ranges from 0 (random) −100 (perfect regularity) using the formula:

$$SRI = -100 + \frac{200}{M(N-1)} \sum_{j=1}^{M} \sum_{i=1}^{N} \delta\left(s_{i,j}, s_{i+1,j}\right)$$

where M = number of daily epochs; N = number of days; $s_{i,j} = 0$ for sleep and 1 for wake; and $\delta\left(s_{i,j}, s_{i+1,j}\right) = 1$ when $s_{i,j} = s_{i+1,j}$ and 0 when $s_{i,j} \neq s_{i+1,j}$. SRI was calculated using the sleepreg R-package (Windred, Stone, et al., 2021). A binary sleep-wake format was used, with diurnal shifts in the sleep/wake cycle assessed using the actigraphy-derived sleep onset and offset times.

Participants were split by napping status for further analyses to establish the relationship with SRI scores. Twelve participants napped at least once in the study duration and were considered 'nappers', with the remaining nine considered 'non-nappers'

Statistical analysis

The distribution of sleep onset, sleep offset, WASO, SE and SOL were considered non-normal (Shapiro-Wilk, $p < .05$). SRI scores were also treated as non-normal due to negative skew, confirmed through visual inspection of Q-Q plots and in alignment with previous research (Windred, Jones, et al., 2021). Descriptive analyses of sleep parameters are presented as median (IQR). Differences between nights preceding morning training and free mornings were assessed using Wilcoxon signed-rank tests. Differences in sleep parameters across days of the

week were evaluated with Friedman tests, and significant results underwent post-hoc analysis using Wilcoxon signed-rank tests for each day-pair, with Bonferroni correction applied for multiple tests to determine *p*-value significance. A Mann-Whitney U test was performed to compare SRI between napper and non-napper groups. Pearson bivariate correlations assessed the relationship between SRI and other sleep parameters, with bias-corrected and accelerated 95% confidence intervals (1000 samples) reported. Data analysis was performed using R (v4.1.3) and SPSS (v29.0.0) statistical software.

Results

Athlete sleep screening questionnaire

The mean ASSQ-SDS was 6 ± 2, with players categorised into none ($n = 5$), mild ($n = 13$) and moderate ($n = 3$) sleep problem groups. No participants presented with severe sleep problems. The mean ASSQ-Chronotype score was 7 ± 2, with one student-athlete classed as having an extreme evening chronotype (ASSQ-Chronotype ≤4).

Sleep parameters

Table 2 presents actigraphy-derived nocturnal sleep parameters across all nights. At the person-level, 14/21 (66.7%) displayed a mean TST of <7 hours per night. For the napping subgroup ($n = 12$), The median nap TST was 1.91 hr (1.28 hr) with a nap onset of 13:28 (02:33) and a nap offset of 15:11 (03:30). The median cumulative sleep duration was 6.90 hr (2.44 hr), with 9/21 (42.9%) participants displaying a mean cumulative sleep duration of <7 hours across the study.

A statistically significant difference in cumulative sleep duration was observed based on the day of the week (Friedman, $\chi^2(6) = 51.531$, $p < .001$; Figure 1(a)). The post-hoc analysis identified significant differences in eight day-pairs. Furthermore, both sleep onset ($\chi^2(6) = 32.857$, $p < .001$) and sleep offset ($\chi^2(6) = 62.571$, $p < .001$) times significantly varied depending on the day of the week, with significant post-hoc differences observed between 7 and 12 day-pairs for onset and offset respectively (Figure 1(b)). Additional significant differences were found between eight day-pairs for the sleep period ($\chi^2(6) = 29.036$, $p < .001$), nine day-pairs for TST ($\chi^2(6) = 29.036$, $p < .001$), and four day-pairs for WASO ($\chi^2(6) = 18.429$, $p = .005$). No significant day-pair differences were identified for SE following post-hoc analysis ($\chi2(6) = 13.250$, $p = .039$) and SOL ($\chi2(6) = 1.433$, $p = .963$). Complete results for post-hoc analysis assessing day-pair differences are presented in Table 3.

Among the seven examined nocturnal sleep parameters, 37/41 (90.2%) day-pairs with a significant difference occurred between a night preceding morning training (Monday, Thursday, and Sunday) and a night preceding free mornings (Tuesday, Wednesday, Friday, and Saturday). When results are grouped accordingly, nights preceding morning training exhibited a shorter sleep period (2.13 hr), shorter TST (1.80 hr), shorter WASO (0.26 hr), earlier sleep onset (00:53), and earlier sleep offset (03:12) compared to nights preceding free mornings (Wilcoxon, all $p < .001$; Table 2). No significant differences were observed between groups for SE (1.5% increase preceding morning training, $p = .053$). In total, 93/109 (85.3%) of nights preceding morning training had a TST <7 hr, compared to 41/121 (33.8%) of nights preceding free mornings.

Sleep regularity

The student-athletes exhibited a median (IQR) SRI of 67.0 (17.0), with scores ranging between 38.1 and 79.8. Participants who napped had a significantly lower median SRI of 57.9 (15.9) compared to a median of 73.6 (9.1) in non-nappers (Mann-Whitney, $z = -2.88$, $p = .004$; Figure 2(a)).

Pearson bivariate correlations revealed an inverse correlation between SRI and cumulative sleep duration ($r = -.62$, $p = .003$, 95%CI [−.79, −.44]; Figure 2(b)). However, when considering only nocturnal sleep, TST was not associated with SRI ($r = -.01$, $p = .949$, 95%CI [−.64, .54]). Lower SRI scores were also significantly inversely associated with a delayed sleep onset ($r = -.50$, $p = .022$, 95%CI [−.75, −.03]) and sleep offset ($r = -.60$, $p = .004$, 95%CI [−.82, −.37]), but not significantly associated with sleep period ($r = .10$, $p = .662$, 95%CI [−.52, .59]), SE ($r = -.34$, $p = .134$, 95%CI [−.71, .12]), WASO ($r = .33$, $p = .147$, 95%CI [−.13, .65]), or SOL ($r = -.17$, $p = .453$, 95%CI [−.59, .38]). With respect to the ASSQ, SRI was positively associated with ASSQ-Chronotype score ($r = .52$, $p = .016$, 95%CI [.02, .81]), indicating lower SRI scores in student-athletes with more evening-type chronotypes. ASSQ-SDS was not associated with SRI ($r = -.31$, $p = .172$, 95%CI [−.57, .02])).

Discussion

The purpose of this study was to examine sleep/wake regularity in student-athletes that are exposed to early morning training.

Table 2. Nocturnal sleep parameters for all nights, and split by nights preceding morning training and preceding free mornings.

Variable	All nights ($n = 230$)	Nights preceding morning training ($n = 109$)	Nights preceding free mornings ($n = 121$)	Morning training vs. free mornings effect size *(r)*
Sleep period (hr)	7.48 (2.59)	6.73 (1.37)	8.86 (2.06)	.69 (large)*
TST (hr)	6.67 (2.19)	5.86 (1.32)	7.66 (1.85)	.67 (large)*
WASO (hr)	0.85 (0.65)	0.72 (0.42)	0.98 (0.72)	.49 (medium)*
SE (%)	88.7 (6.7)	89.6 (6.1)	88.1 (7.1)	.19 (small)
Sleep onset (hh:mm)	23:37 (01:40)	23:07 (01:21)	00:00 (01:50)	.50 (large)*
Sleep offset (hh:mm)	07:08 (03:17)	05:45 (00:30)	08:57 (01:32)	.85 (large)*
SOL (hr)	0.08 (0.10)	0.08 (0.09)	0.08 (0.10)	.08 (negligible)

Data presented as median (interquartile range). Abbreviations: TST = total sleep time; WASO = wake after sleep onset; SE = sleep efficiency; SOL = sleep onset latency. *Denotes significant difference between groups ($p < .05$). Effect sizes (*r*) interpreted using guidelines by Cohen (1988).

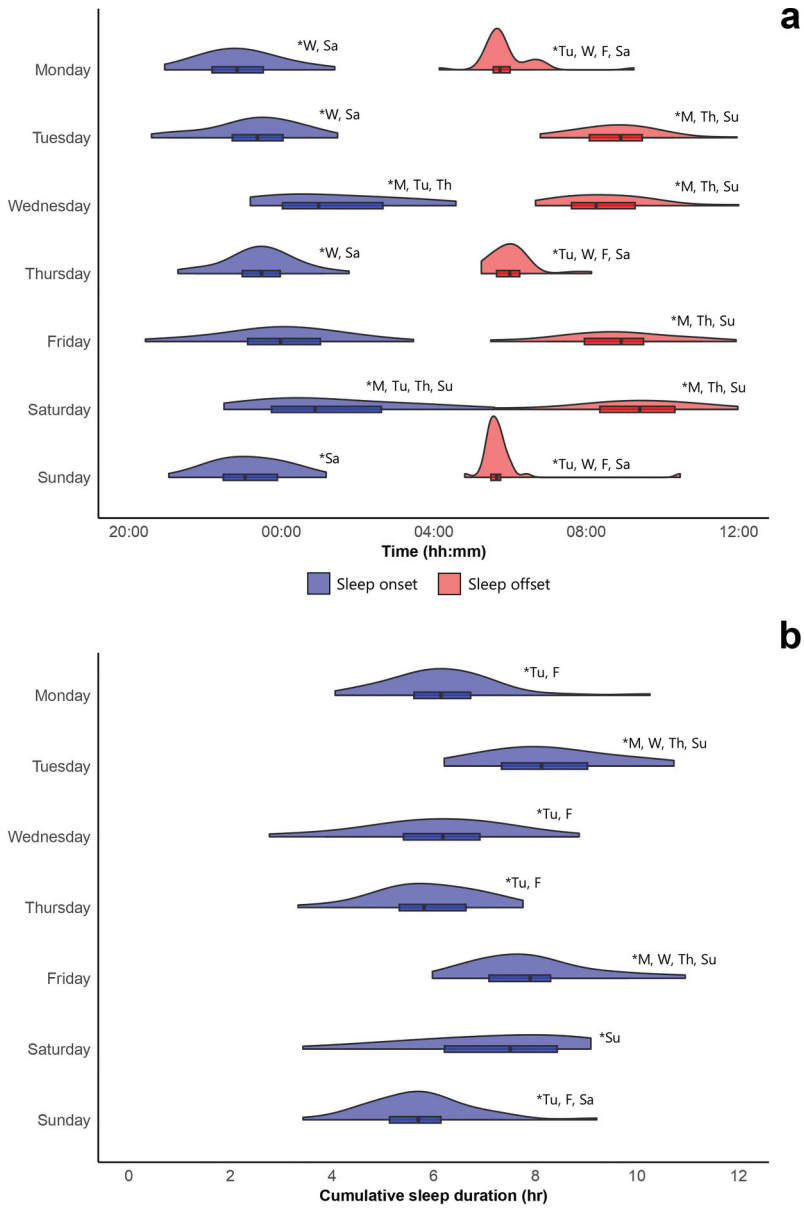

Figure 1. Violin plots demonstrating the distribution of **(A)** nocturnal sleep onset and offset times, split by day of the week, and **(B)** cumulative sleep duration including napping *denotes statistically significant difference in post-hoc Wilcoxon rank-sum test with Bonferroni adjustment applied to the *p*-value for multiple tests ($p < .00238$). Abbreviations: *M* = Monday; Tu = Tuesday; W = Wednesday; Th = Thursday; F = Friday; Sa = Saturday; Su = Sunday.

When assessed using the SRI, student-athletes displayed irregular sleep/wake patterns in response to changing daily training demands, with nights preceding morning training exhibiting advanced sleep onset and offset times, and shorter sleep durations. Additionally, SRI was significantly lower in student-athletes that napped compared to non-nappers, and a lower SRI was significantly associated with greater eveningness in chronotype assessed using the ASSQ, increased cumulative sleep duration, and delayed sleep onset and offset times.

The sleep parameters highlight a clear disparity between nights preceding morning training and free mornings. This observation aligns with previous findings in NCAA student-athletes that showed a significant reduction in TST on nights preceding morning training (Merfeld et al., 2022). While student-athletes attempted to compensate for the early start enforced by training by advancing sleep onset by 0.88 hours on nights preceding morning training, this adjustment did not fully compensate for the advance in sleep offset, which exceeded 3 hours. To fully compensate for morning training, a student-athlete wishing to wake 05:30 in advance of training at 06:30 and achieve a sleep period of 8 hours would need to fall asleep by 21:30. However, this may not be feasible due to both homeostatic (insufficient sleep pressure) and circadian (high alertness) processes at this time (Borbély et al., 2016), in addition to any academic, social, or work demands in the late evening. With respect to sleep durations, both the median TST of 6.67 hours and cumulative sleep duration of 6.90 hours fall below public health recommendations for young adults (Hirshkowitz et al., 2015), indicating that many student-athletes may not be meeting their optimal sleep needs. The notable differences in sleep parameters between days reflect the unique scheduling of the student-athlete day, which

Table 3. Post-hoc analysis of Wilcoxon signed-rank tests to assess between-day differences in actigraphy-derived sleep outcomes.

Day-pair	Sleep period	TST	WASO	SE	Sleep onset	Sleep offset	SOL	Cumulative sleep duration
Mon-Tue	−4.924*	−4.895*	−4.155*	−1.617	−1.777	−5.373*	−.966	−4.895*
Mon-Wed	−.678	−.330	−2.381	−2.902	−3.632*	−4.015*	−.141	−.330
Mon-Thu	−1.410	−1.205	−.966	−.470	−2.317	−1.325	−1.244	−1.205
Mon-Fri	−4.390*	−4.311*	−2.704	−.901	−2.948	−5.110*	−.761	−4.422*
Mon-Sat	−2.787	−2.138	−1.970	−1.129	−4.373*	−4.493*	−.757	−2.883
Mon-Sun	−1.523	−1.911	−.598	−1.065	−1.359	−1.599	−1.178	−1.507
Tue-Wed	−3.424*	−3.493*	−.817	−.852	−3.354*	−.678	−.465	−3.493*
Tue-Thu	−5.086*	−5.018*	−4.351*	−2.146	−.607	−5.086*	−.422	−5.018*
Tue-Fri	−1.245	−.638	−1.416	−1.330	−2.228	−.721	−.154	−1.179
Tue-Sat	−2.595	−2.571	−1.441	−.553	−4.036*	−1.802	−.795	−2.475
Tue-Sun	−5.143*	−4.595*	−3.063*	−.377	−.280	−4.918*	−.286	−5.143*
Wed-Thu	−1.894	−.539	−3.007	−3.007	−3.945*	−4.015*	−.310	−.539
Wed-Fri	−2.972	−2.510	−.391	−1.569	−2.450	−1.025	−.653	−3.215*
Wed-Sat	−1.929	−2.275	−3.92	−.941	−.156	−2.624	−.227	−2.381
Wed-Sun	−1.721	−.392	−1.964	−1.547	−3.010	−3.294*	−.259	−1.199
Thu-Fri	−4.796*	−4.292*	−3.377*	−1.655	−1.188	−4.967*	−.392	−4.710*
Thu-Sat	−2.739	−3.162*	−2.866	1.816	−3.556*	−4.397*	−.127	−2.811
Thu-Sun	−.231	−.874	−1.940	−2.111	−1.184	−2.915	−.075	−.812
Fri-Sat	−1.730	−1.874	−.312	−.793	−2.402	−1.257	−.162	−1.874
Fri-Sun	−4.783*	−4.717*	−1.765	−.648	−2.065	−4.444*	−.661	−4.717*
Sat-Sun	−3.147*	−3.123*	−1.689	−.317	−3.111*	−4.015*	−.200	−3.123*

Data presented as Wilcoxon signed-rank Z-score. Abbreviations: TST = total sleep time; WASO = wake after sleep onset; SE = sleep efficiency; Mon = Monday; Tue = Tuesday; Wed = Wednesday; Thu = Thursday; Fri = Friday; Sat = Saturday; Sun = Sunday.
*Denotes significant difference between day-pair at Bonferroni-corrected significance level ($p < .00238$).

involves sport training and competition distributed throughout the week, along with academic and social demands (Brauer et al., 2019). This contrasts with a more typical working schedule where demands are consistent over multiple consecutive days, creating a sleep pattern resembling social jetlag on a daily basis in student-athletes. These increased levels of variability can prevent the establishment of consistent sleep/wake patterns, impacting healthy circadian regulation (Wittmann et al., 2006).

The SRI scores observed indicate that the student-athletes in this study exhibit some of the most irregular sleep/wake patterns among healthy populations. All participants had SRI scores below the median of 81.0 observed in the British population using data from the UK Biobank (Windred, Jones, et al.,

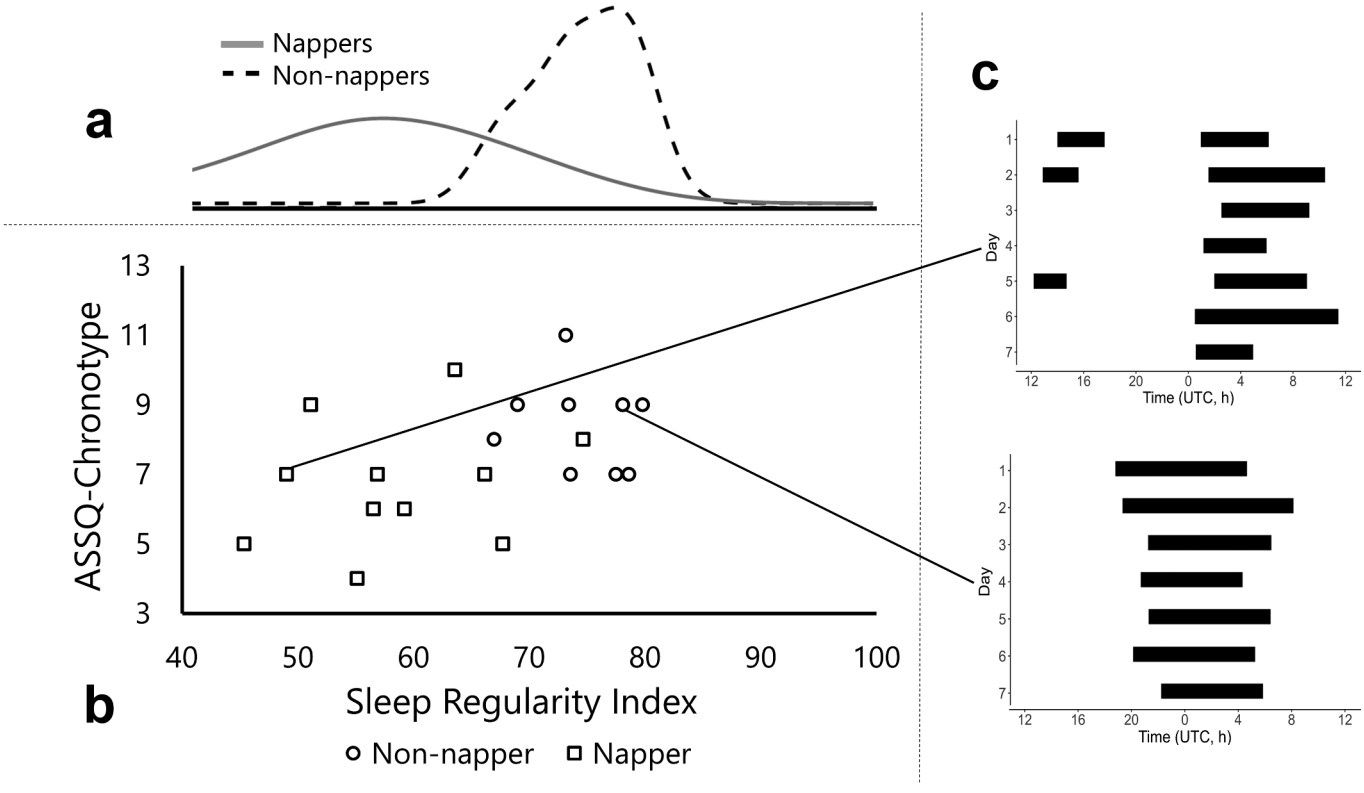

Figure 2. (a) Smoothed kernel density estimation of SRI student-athletes in the study classified as nappers and non-nappers; (b) Scatter plot highlighting the positive relationship between SRI and ASSQ-Chronotype, with data points split by napping status; (c) Example raster plots of two participants displaying differing SRI scores and napping status.

2021), whilst 9/21 participants (42.9%) presented an SRI that would place them in the lowest 10th percentile (SRI <65.1). Previous studies on university students in Australia (mean SRI = 79.9) (Windred, Stone, et al., 2021) and the United States (mean SRI = 73) (Phillips et al., 2017) also demonstrated greater sleep/wake regularity. The low SRI observed also aligns with previous findings by Leduc et al. (2020), demonstrating greater sleep variability in British student-athletes compared to non-athlete peers (Leduc et al., 2020), suggesting that the combination of performance sport and academic study can lead to sleep/wake irregularity. The SRI in this study was notably lower than observed by Halson et al. (2022) in a cohort of elite athletes (median SRI = 85.1) and by Teece et al. (2024) in professional rugby union players (median SRI = 76.4), although only nocturnal sleep episodes were considered in these studies However, Alves Facundo et al. (2022) demonstrated lower SRI scores in national (mean SRI = 68.7) and regional (mean SRI = 73.8) level athletes compared to international athletes (mean SRI = 80.1)). The difference between groups may be attributed to the additional demands on sub-elite athletes, including work and/or education commitments, as is often the case for student-athletes (Brauer et al., 2019). Additionally, lower SRI values were found in team sports compared to individual sports and in male athletes compared to females (Alves Facundo et al., 2022), providing further rationale for the irregular sleep/wake patterns present within this study.

When considering the relationship between SRI and sleep parameters, we observed an inverse relationship between SRI and sleep onset, sleep offset, and cumulative sleep duration, while participants classified as nappers had a lower SRI than non-nappers. However, no relationship was found between TST and SRI. This finding partially contrasts with previous research by Phillips et al. (2017), which showed greater daytime sleep and delayed sleep timing in irregular sleepers but found no relationship between cumulative sleep duration and SRI . Therefore, it appears that naps were used to add to nocturnal sleep, rather than compensate for lower nocturnal sleep than non-napping peers. However, this practice comes with the consequence of lower sleep regularity due to the variable placement and duration of naps during the day. It is unclear whether the potential benefit of increasing cumulative sleep duration through napping – in a population where sleep durations often fall below public health recommendations (Hirshkowitz et al., 2015) – outweigh any negative effects of the subsequent reduction in sleep regularity and requires attention in future research. In addition, while shorter naps of 10–20 minutes are recommended for adults with regular sleep/wake schedules (Milner & Cote, 2009), the longer naps observed in our study may result from restricted nocturnal sleep opportunity, as demonstrated by the advanced sleep offset times and shortened TST on nights preceding morning training.

SRI was also significantly associated with ASSQ-Chronotype, revealing that a later chronotype was associated with greater sleep/wake irregularity. While student-athletes may generally display a greater tendency towards early chronotypes than non-athlete students (Litwic-Kaminska & Kotysko, 2020), age-related changes in chronotype mean that a substantial number of student-athletes will still present with later chronotypes (Roenneberg et al., 2004). Although the ASSQ-Chronotype does not enable grouping of participants, the results do not indicate any skew towards morning-types, in contrast to other elite athlete groups that train in the morning in response to adaptation to early morning starts or being drawn towards sports that match their own chronotype (Lastella et al., 2016; Rae et al., 2015). The impact of aligning the timing of sport training and competition to circadian preferences is currently unclear. However, current evidence suggests some favourable outcomes, such as reduced perceived exertion and improved performance on standardised physical tests (Lastella et al., 2021; Roveda et al., 2020).

It is evident from our results that the observed sleep/wake irregularity in student-athletes has important implications and necessitates additional consideration. At a population level, greater variation in sleep timing has been associated with various health concerns, including physical, mental, and metabolic health (Sletten et al., 2023), in addition to all-cause mortality (Omichi et al., 2022). Furthermore, research by Phillips et al. (2017) found a relationship between lower SRI scores and academic performance , possibly linked to experimental findings showing greater morning sleepiness and heightened negative mood states in the evenings when sleep variability is high (Sun et al., 2023). However, it is currently unclear how SRI is related to components of sports performance. Our findings underscore the importance of future research exploring sleep in student-athletes to consider sleep across the 24-hour day and utilise measures of sleep variability (Kemp et al., 2023; Nedelec et al., 2018).

There is a need to explore changes that can be made at the individual, social, and societal levels to support regular sleep in student-athletes (Grandner, 2019). For instance, a previous sleep workshop intervention that allowed student-athletes to self-select a sleep behaviour to change showed that maintenance of a more consistent sleep/wake cycle was the most common selection (Kaier et al., 2016). Future studies should utilise behaviour change theory to identify barriers to achieving consistent sleep (Wilson et al., 2024). However, achieving a meaningful change may be challenging with the continued scheduling of training in the morning, and therefore a more effective social-level change could be to adjust training timings to align with age-related changes in chronotype. Delayed start times in school settings have proven effective in improving sleep and classroom performance (Watson et al., 2017), prompting the need to explore similar changes among student-athletes.

There are limitations to the present study that should be considered. Firstly, academic scheduling was not considered as lessons on all days except Thursday were preceded by sports training. However, earlier class start times have been associated with unfavourable changes to sleep/wake behaviours and poorer academic attainment (Yeo et al., 2023), and as such could have influenced sleep on a Wednesday night. Secondly, all student-athletes resided off-campus during the study. Travelling to the training venue likely enforced an earlier sleep offset time than may have been present if residing on-site, and a comparison between accommodation locations warrants future research. Thirdly, the timing and location of Wednesday matches differed between the two weeks. This did not result in any

significant change in sleep parameters between subsequent nights, although greater discrepancies in competition timing are present throughout the season and can influence sleep parameters in athletes (Sargent & Roach, 2016). Lastly, the study results may not be fully representative of the entire student-athlete population. For instance, student-athletes who train in the evening after lessons may align better with age-related changes in chronotype (Roenneberg et al., 2004), whilst between-sex chronobiological differences mean that morning training is less disruptive to sleep in female groups (Power et al., 2023). Therefore, future research should examine sleep/wake regularity in female student-athlete groups, and the influence of academic scheduling in student-athlete groups that do not have training sessions prior to lessons.

Conclusion

The current study has been the first to assess sleep/wake regularity using the SRI in a student-athlete population and revealed that student-athletes exposed to early morning training exhibited irregular sleep/wake patterns and undesirable sleep parameters on nights preceding training. These findings carry implications for both sports and academic performance, as well as for physical and mental health. Stakeholders in sports should carefully consider the timing of training and matches for student-athletes and take into account the impact these structural factors have on sleep. Further research is warranted to explore the potential implications of irregular sleep/wake timing on circadian health and to investigate novel interventions targeted specifically at improving sleep regularity in student-athletes.

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

Alves Facundo, L., Brant, V. M., Guerreiro, R. C., Andrade, H. D. A., Louzada, F. M., Silva, A., & Mello, M. T. D. (2022). Sleep regularity in athletes: Comparing sex, competitive level and sport type. *Chronobiology International*, *39*(10), 1381–1388. https://doi.org/10.1080/07420528.2022.2108716

Bender, A. M., Lawson, D., Werthner, P., & Samuels, C. H. (2018). The clinical validation of the Athlete sleep screening questionnaire: An instrument to identify athletes that need further sleep assessment. *Sports Medicine - Open*, *4*(1), 23. https://doi.org/10.1186/s40798-018-0140-5

Benjamin, C. L., Adams, W. M., Curtis, R. M., Sekiguchi, Y., Giersch, G. E. W., & Casa, D. J. (2019). Early morning training impacts previous night's sleep in NCAA Division I cross country runners. *Women in Sport & Physical Activity Journal*, *28*(1), 27–33. https://doi.org/10.1123/wspaj.2018-0050

Borbély, A. A., Daan, S., Wirz-Justice, A., & Deboer, T. (2016). The two-process model of sleep regulation: A reappraisal. *Journal of Sleep Research*, *25*(2), 131–143. https://doi.org/10.1111/jsr.12371

Brauer, A. A., Athey, A. B., Ross, M. J., & Grandner, M. A. (2019). Sleep and health among collegiate student athletes. *Chest*, *156*(6), 1234–1245. https://doi.org/10.1016/j.chest.2019.08.1921

Caia, J., Halson, S. L., Scott, T. J., & Kelly, V. G. (2017). Intra-individual variability in the sleep of senior and junior rugby league athletes during the competitive season. *Chronobiology International*, *34*(9), 1239–1247. https://doi.org/10.1080/07420528.2017.1358736

Carney, C. E., Buysse, D. J., Ancoli-Israel, S., Edinger, J. D., Krystal, A. D., Lichstein, K. L., & Morin, C. M. (2012). The consensus sleep diary: Standardizing prospective sleep self-monitoring. *Sleep: Journal of Sleep Research & Sleep Medicine*, *35*(2), 287–302. https://doi.org/10.5665/sleep.1642

Carter, J. R., Gervais, B. M., Adomeit, J. L., & Greenlund, I. M. (2020). Subjective and objective sleep differ in male and female collegiate athletes. *Sleep Health*, *6*(5), 623–628. https://doi.org/10.1016/j.sleh.2020.01.016

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Routledge. https://doi.org/10.4324/9780203771587.

Driller, M. W., Dixon, Z. T., & Clark, M. I. (2017). Accelerometer-based sleep behavior and activity levels in student athletes in comparison to student non-athletes. *Sport Sciences for Health*, *13*(2), 411–418. https://doi.org/10.1007/s11332-017-0373-6

Fischer, D., Klerman, E. B., & Phillips, A. J. K. (2021). Measuring sleep regularity: Theoretical properties and practical usage of existing metrics. *Sleep: Journal of Sleep Research & Sleep Medicine*, *44*(10). https://doi.org/10.1093/sleep/zsab103

Grandner, M. A. (2019). Social-ecological model of sleep health. In M. Grandner (Ed.), *Sleep and health* (pp. 45–53). Academic Press. https://doi.org/10.1016/B978-0-12-815373-4.00005-8

Halson, S. L., Johnston, R. D., Piromalli, L., Lalor, B. J., Cormack, S., Roach, G. D., & Sargent, C. (2022). Sleep regularity and predictors of sleep efficiency and sleep duration in elite team sport athletes. *Sports Medicine - Open*, *8*(1), 79. https://doi.org/10.1186/s40798-022-00470-7

Hirshkowitz, M., Whiton, K., Albert, S. M., Alessi, C., Bruni, O., DonCarlos, L., Hazen, N., Herman, J., Katz, E. S., Kheirandish-Gozal, L., Neubauer, D. N., O'Donnell, A. E., Ohayon, M., Peever, J., Rawding, R., Sachdeva, R. C., Setters, B., Vitiello, M. V., Ware, J. C., & Adams Hillard, P. J. (2015). National Sleep Foundation's sleep time duration recommendations: Methodology and results summary. *Sleep Health*, *1*(1), 40–43. https://doi.org/10.1016/j.sleh.2014.12.010

Kaier, E., Zanotti, D., Davis, J. L., Strunk, K., & Cromer, L. D. (2016). Addressing the problem of student-athlete sleepiness: Feasibility of implementing an interactive sleep workshop at a Division I school. *Journal of Clinical Sport Psychology*, *10*(3), 237–247. https://doi.org/10.1123/jcsp.2015-0023

Kemp, S., Spence, A. L., Keller, B. S., Ducker, K. J., & Gucciardi, D. F. (2023). Intraindividual variability in sleep among athletes: A systematic review of definitions, operationalizations, and key correlates. *Scandinavian Journal of Medicine & Science in Sports*, *33*(12), 2413–2422. https://doi.org/10.1111/sms.14453

Kroshus, E., Wagner, J., Wyrick, D., Athey, A., Bell, L., Benjamin, H. J., Grandner, M. A., Kline, C. E., Mohler, J. M., Roxanne Prichard, J., Watson, N. F., & Hainline, B. (2019). Wake up call for collegiate athlete

sleep: Narrative review and consensus recommendations from the NCAA interassociation task force on sleep and wellness. *British Journal of Sports Medicine*, *53*(12), 731–736. https://doi.org/10.1136/bjsports-2019-100590

Lastella, M., Miller, D. J., Quilelli, M., Roberts, S., Aisbett, B., & Condo, D. (2021). The impact of chronotype on the sleep and training responses of elite female Australian footballers. *Clocks & Sleep*, *3*(4), 528–535. https://doi.org/10.3390/clockssleep3040037

Lastella, M., Roach, G. D., Halson, S. L., & Sargent, C. (2016). The chronotype of elite athletes. *Journal of Human Kinetics*, *54*(1), 219–225. https://doi.org/10.1515/hukin-2016-0049

Leduc, C., Tee, J., Weakley, J., Ramirez, C., & Jones, B. (2020). The quality, quantity, and intraindividual variability of sleep among students and student-athletes. *Sports Health*, *12*(1), 43–50. https://doi.org/10.1177/1941738119887966

Litwic-Kaminska, K., & Kotysko, M. (2020). Sleep quality of student athletes and non-athletes - the role of chronotype, stress and life satisfaction. *Sleep Science*, *13*(4), 249–255. https://doi.org/10.5935/1984-0063.20190153

Lund, H. G., Reider, B. D., Whiting, A. B., & Prichard, J. R. (2010). Sleep patterns and predictors of disturbed sleep in a large population of college students. *Journal of Adolescent Health*, *46*(2), 124–132. https://doi.org/10.1016/j.jadohealth.2009.06.016

Mah, C. D., Kezirian, E. J., Marcello, B. M., & Dement, W. C. (2018). Poor sleep quality and insufficient sleep of a collegiate student-athlete population. *Sleep Health*, *4*(3), 251–257. https://doi.org/10.1016/j.sleh.2018.02.005

Merfeld, B., Mancosky, A., Luedke, J., Griesmer, S., Erickson, J. L., & Carvalho, V. (2022). The impact of early morning training sessions on total sleep time in collegiate athletes. *International Journal of Exercise Science*, *15*(6), 423–433. https://www.ncbi.nlm.nih.gov/pubmed/35519439

Migueles, J. H., Rowlands, A. V., Huber, F., Sabia, S., & Van Hees, V. T. (2019). GGIR: A research community-driven open source R package for generating physical activity and sleep outcomes from multi-day raw accelerometer data. *Journal for the Measurement of Physical Behaviour*, *2*(3), 188–196. https://doi.org/10.1123/jmpb.2018-0063

Milner, C. E., & Cote, K. A. (2009). Benefits of napping in healthy adults: Impact of nap length, time of day, age, and experience with napping. *Journal of Sleep Research*, *18*(2), 272–281. https://doi.org/10.1111/j.1365-2869.2008.00718.x

Nedelec, M., Aloulou, A., Duforez, F., Meyer, T., & Dupont, G. (2018). The variability of sleep among elite athletes. *Sports Medicine - Open*, *4*(1), 34. https://doi.org/10.1186/s40798-018-0151-2

Omichi, C., Koyama, T., Kadotani, H., Ozaki, E., Tomida, S., Yoshida, T., Otonari, J., Ikezaki, H., Hara, M., Tanaka, K., Tamura, T., Nagayoshi, M., Okada, R., Kubo, Y., Oze, I., Matsuo, K., Nakamura, Y., Kusakabe, M., . . . Takeuchi, K. (2022). Irregular sleep and all-cause mortality: A large prospective cohort study. *Sleep Health*, *8*(6), 678–683. https://doi.org/10.1016/j.sleh.2022.08.010

Phillips, A. J. K., Clerx, W. M., O'Brien, C. S., Sano, A., Barger, L. K., Picard, R. W., Lockley, S. W., Klerman, E. B., & Czeisler, C. A. (2017). Irregular sleep/wake patterns are associated with poorer academic performance and delayed circadian and sleep/wake timing. *Scientific Reports*, *7*(1). https://doi.org/10.1038/s41598-017-03171-4

Plekhanova, T., Rowlands, A. V., Yates, T., Hall, A., Brady, E. M., Davies, M., Khunti, K., & Edwardson, C. L. (2020). Equivalency of sleep estimates: Comparison of three research-grade accelerometers. *Journal for the Measurement of Physical Behaviour*, *3*(4), 294–303. https://doi.org/10.1123/jmpb.2019-0047

Power, C. J., Fox, J. L., Elliott-Sale, K. J., Bender, A. M., Dalbo, V. J., & Scanlan, A. T. (2023). Waking up to the issue! Research inattention and sex-related differences warrant more sleep studies in female athletes. *Sports Medicine*, 1–19. https://doi.org/10.1007/s40279-023-01963-50

Rabin, J. M., Mehra, R., Chen, E., Ahmadi, R., Jin, Y., & Day, C. (2020). Assessment of sleep health in collegiate athletes using the Athlete sleep screening questionnaire. *Journal of Clinical Sleep Medicine*, *16*(8), 1349–1356. https://doi.org/10.5664/jcsm.8532

Rae, D. E., Stephenson, K. J., & Roden, L. C. (2015). Factors to consider when assessing diurnal variation in sports performance: The influence of chronotype and habitual training time-of-day. *European Journal of Applied Physiology*, *115*(6), 1339–1349. https://doi.org/10.1007/s00421-015-3109-9

Rebello, L. J., Roberts, A. W., Fenuta, A. M., Cote, A. T., & Bodner, M. E. (2022). Sleep quality and sleep behaviors in varsity athletes: A pilot study. *Frontiers in Sports and Active Living*, *4*, 906663. https://doi.org/10.3389/fspor.2022.906663

Roenneberg, T., Kuehnle, T., Pramstaller, P. P., Ricken, J., Havel, M., Guth, A., & Merrow, M. (2004). A marker for the end of adolescence. *Current Biology*, *14*(24), R1038–9. https://doi.org/10.1016/j.cub.2004.11.039

Roveda, E., Mulè, A., Galasso, L., Castelli, L., Scurati, R., Michielon, G., Esposito, F., Caumo, A., & Montaruli, A. (2020). Effect of chronotype on motor skills specific to soccer in adolescent players. *Chronobiology International*, *37*(4), 552–563. https://doi.org/10.1080/07420528.2020.1729787

Samuels, C., James, L., Lawson, D., & Meeuwisse, W. (2016). The Athlete sleep screening questionnaire: A new tool for assessing and managing sleep in elite athletes. *British Journal of Sports Medicine*, *50*(7), 418–422. https://doi.org/10.1136/bjsports-2014-094332

Sargent, C., Lastella, M., Halson, S. L., & Roach, G. D. (2014). The impact of training schedules on the sleep and fatigue of elite athletes. *Chronobiology International*, *31*(10), 1160–1168. https://doi.org/10.3109/07420528.2014.957306

Sargent, C., & Roach, G. (2016). Sleep duration is reduced in elite athletes following night-time competition. *Chronobiology International*, *33*(6), 667–670. https://doi.org/10.3109/07420528.2016.1167715

Sletten, T. L., Weaver, M. D., Foster, R. G., Gozal, D., Klerman, E. B., Rajaratnam, S. M. W., Roenneberg, T., Takahashi, J. S., Turek, F. W., Vitiello, M. V., Young, M. W., & Czeisler, C. A. (2023). The importance of sleep regularity: A consensus statement of the national sleep foundation sleep timing and variability panel. *Sleep Health*, *9*(6), 801–820. https://doi.org/10.1016/j.sleh.2023.07.016

Stephenson, K. L., Trbovich, A. M., Vandermark, L. W., McDermott, B. P., Henry, L. C., Anderson, M. N., & Elbin, R. J. (2022). Exploring the effect of napping on sleep quality and duration in collegiate athletes. *Journal of American College Health*, *70*(5), 1451–1456. https://doi.org/10.1080/07448481.2020.1803881

Sun, W., Cheung, F. T. W., Chan, N. Y., Zhang, J., Chan, J. W. Y., Chan, K. C. C., Wing, Y. K., & Li, S. X. (2023). The impacts of intra-individual daily sleep variability on daytime functioning and sleep architecture in healthy young adults: An experimental study. *Journal of Sleep Research*, *33*(3). https://doi.org/10.1111/jsr.13967

Teece, A., Beaven, C. M., Suppiah, H., Argus, C. K., Gill, N., & Driller, M. W. (2024). Routine, routine, routine: Sleep regularity and its association with sleep metrics in professional rugby union athletes. *Sports Medicine - Open*, *10*(1), 51. https://doi.org/10.1186/s40798-024-00709-5

Van Hees, V. T., Gorzelniak, L., Dean León, E. C., Eder, M., Pias, M., Taherian, S., Ekelund, U., Renström, F., Franks, P. W., Horsch, A., & Brage, S. (2013). Separating movement and gravity components in an acceleration signal and implications for the assessment of human daily physical activity. *PLOS ONE*, *8*(4), e61691. https://doi.org/10.1371/journal.pone.0061691

Van Hees, V. T., Sabia, S., Anderson, K. N., Denton, S. J., Oliver, J., Catt, M., Abell, J. G., Kivimäki, M., Trenell, M. I., & Singh-Manoux, A. (2015). A novel, open access method to assess sleep duration using a wrist-worn accelerometer. *PLOS ONE*, *10*(11), e0142533. https://doi.org/10.1371/journal.pone.0142533

Walsh, N. P., Halson, S. L., Sargent, C., Roach, G. D., Nedelec, M., Gupta, L., Leeder, J., Fullagar, H. H., Coutts, A. J., Edwards, B. J., Pullinger, S. A., Robertson, C. M., Burniston, J. G., Lastella, M., Le Meur, Y., Hausswirth, C., Bender, A. M., Grandner, M. A., & Samuels, C. H. (2021). Sleep and the athlete: Narrative review and 2020 expert consensus recommendations. *British Journal of Sports Medicine*, *55*(7), 356–368. https://doi.org/10.1136/bjsports-2020-102025

Watson, N. F., Martin, J. L., Wise, M. S., Carden, K. A., Kirsch, D. B., Kristo, D. A., Malhotra, R. K., Olson, E. J., Ramar, K., Rosen, I. M., Rowley, J. A., Weaver, T. E., & Chervin, R. D. (2017). Delaying middle school and high school start times promotes student health and performance: An American Academy of Sleep Medicine Position statement. *Journal of*

Clinical Sleep Medicine, *13*(4), 623–625. https://doi.org/10.5664/jcsm.6558

Wilson, S., Draper, S., Jones, M., & Parker, J. (2023). O11 Irregular sleep/wake patterns in student-athletes. *BMJ Open Respiratory Research*, *10*(S1), A6–A7. https://doi.org/10.1136/bmjresp-2023-bssconf.11

Wilson, S. M. B., Sparks, K. V., Cline, A., Draper, S. B., Jones, M. I., & Parker, J. K. (2024). Behavioral interventions and behavior change techniques used to improve sleep outcomes in athlete populations: A scoping review. *Behavioral Sleep Medicine*, *22*(6), 1–23. https://doi.org/10.1080/15402002.2024.2374257

Windred, D. P., Jones, S. E., Russell, A., Burns, A. C., Chan, P., Weedon, M. N., Rutter, M. K., Olivier, P., Vetter, C., Saxena, R., Lane, J. M., Cain, S. W., & Phillips, A. J. K. (2021). Objective assessment of sleep regularity in 60 000 UK Biobank participants using an open-source package. *Sleep: Journal of Sleep Research & Sleep Medicine*, *44*(12). https://doi.org/10.1093/sleep/zsab254

Windred, D. P., Stone, J. E., McGlashan, E., Cain, S. W., & Phillips, A. (2021). Attitudes towards sleep as a time commitment are associated with sleep regularity. *Behavioral Sleep Medicine*, *19*(6), 732–743. https://doi.org/10.1080/15402002.2020.1860989

Wittmann, M., Dinich, J., Merrow, M., & Roenneberg, T. (2006). Social jetlag: Misalignment of biological and social time. *Chronobiology International*, *23*(1–2), 497–509. https://doi.org/10.1080/07420520500545979

Yeo, S. C., Lai, C. K. Y., Tan, J., Lim, S., Chandramoghan, Y., Tan, T. K., & Gooley, J. J. (2023). Early morning university classes are associated with impaired sleep and academic performance. *Nature Human Behaviour*, *7*(4), 502–514. https://doi.org/10.1038/s41562-023-01531-x